

COMPUTATIONS FOR MAZUR'S KNOT AND THE OCTAHEDRON

JACK CALCUT AND YANGYANG DU

ABSTRACT. We provide the code, complete output, and a brief analysis for every computer calculation used in Sections 6 and 7 of our paper *Mazur's Knot and the Octahedron*.

1. INTRODUCTION

This is a companion paper to *Mazur's Knot and the Octahedron* [CD26]. It contains the code, complete output, and a brief analysis for every computer calculation used in Sections 6 and 7 of [CD26]. Together with the GitHub repository, this document makes those computations fully reproducible. All computations were performed using SageMath 10.7, SnapPy 3.3.2, and GAP 4.14.0. The code is available at <https://github.com/yyadu/Mazur-manifold-computation>.

We use the following notation consistent with our paper [CD26]:

K	Mazur knot in $S^1 \times S^2$
X	Mazur knot exterior
μ_K	Oriented meridian of K in ∂X
λ_K	Oriented longitude of K in ∂X
$X(p, q)$	Dehn filling of X using slope $(p, q) \leftrightarrow p/q$
M_k	Mazur 4-manifold with framing $k \in \mathbb{Z}$
∂M_k	Boundary 3-manifold of M_k
J	Jester knot in $S^1 \times S^2$
Y	Jester knot exterior
μ_J	Oriented meridian of J in ∂Y
λ_J	Oriented longitude of J in ∂Y
$Y(p, q)$	Dehn filling of Y using slope $(p, q) \leftrightarrow p/q$
N_k	Jester 4-manifold with framing $k \in \mathbb{Z}$
∂N_k	Boundary 3-manifold of N_k

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2. MAZUR AND JESTER KNOT EXTERIORS

The Mazur knot $K \subset S^1 \times S^2$ and the Jester knot $J \subset S^1 \times S^2$ are shown in Figure 1. Equipping K and J with the framing $k \in \mathbb{Z}$ yields the surgery diagrams for $\partial M_k = X(k, 1)$ and $\partial N_k = Y(k, 1)$ respectively in Figure 1.

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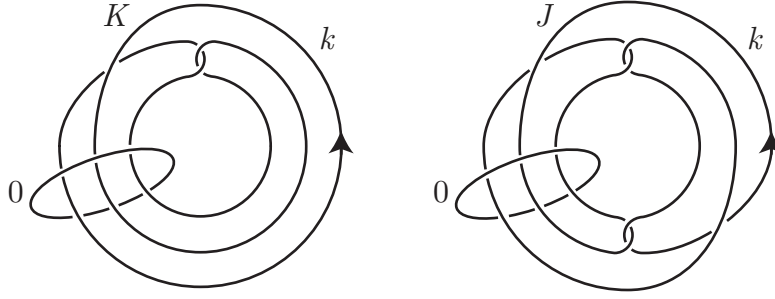


FIGURE 1. Surgery diagrams for $\partial M_k = X(k, 1)$ (left) and $\partial N_k = Y(k, 1)$ (right).

We first import the link diagrams into SnapPy using the PLink editor.

```

1 import snappy
2 # Opens the PLink editor to draw links
3 M = snappy.Manifold()

```

- 1 Starting the link editor.
 - 2 Select Tools->Send to SnapPy to load the link complement.
 - 3 Your new Plink window needs an event loop to become visible.
 - 4 Type "%gui tk" below (without the quotes) to start one.
-

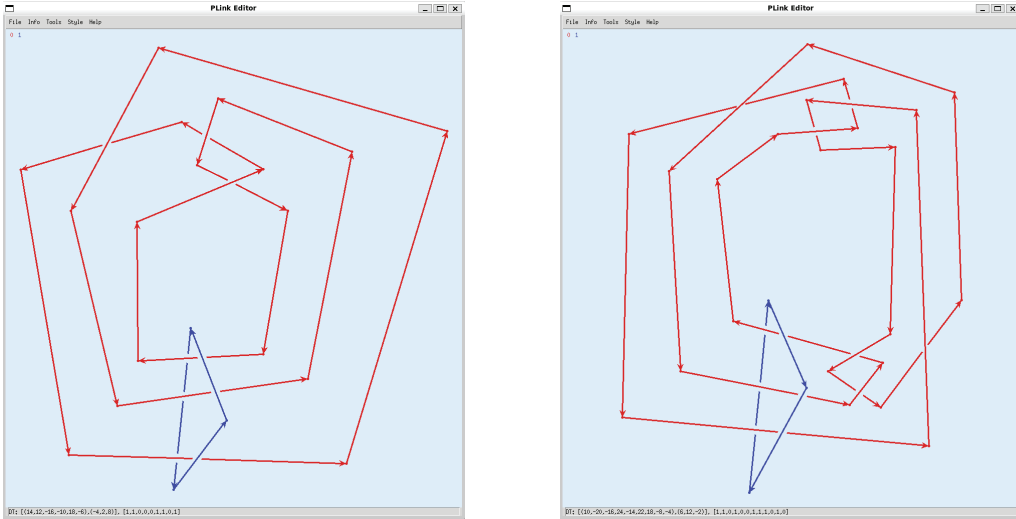


FIGURE 2. PLink Editor link diagrams for Mazur and Jester boundary 3-manifolds, together with their DT codes at the bottom.

The Dowker–Thistlethwaite (DT) codes for the two links in Figure 2 are

- DT: $[(14, 12, -16, -10, 18, -6), (-4, 2, 8)], [1, 1, 0, 0, 0, 1, 1, 0, 1]$
- DT: $[(10, -20, -16, 24, -14, 22, 18, -8, -4), (6, 12, -2)], [1, 1, 0, 1, 0, 0, 1, 1, 1, 0, 1, 0]$

The unknotted blue component is Dehn filled with slope $0/1 \leftrightarrow (0,1)$ to obtain the exterior X of K and Y of J respectively in $S^1 \times S^2$. We compute basic information about X .

```

1 import snappy
2 # Mazur knot exterior X
3 X = snappy.Manifold('DT: [(14,12,-16,-10,18,-6),(-4,2,8)], [1,1,0,0,0,1,1,0,1]')
4 X.dehn_fill((0,1),1)
5 X = X.filled_triangulation()
6 Xid = X.identify()
7 IsomX = X.symmetry_group()
8 print("Mazur knot exterior X")
9 print("Volume =", X.volume())
10 print("Isometric to census manifold:", Xid)
11 print("Cusp shape =", X.cusp_info('shape'))
12 print("")
13 print("Is X achiral?", IsomX.is_amphicheiral())
14 print("Order of isometry group =", IsomX.order())
15 print("Isometry group =", IsomX)
16 print("Self-isometries are:", IsomX.isometries(), "\n")
17 print("Isometries X -> m137 are:", X.is_isometric_to(snappy.Manifold('m137'),
18     return_isometries=True))

```

```

1 Mazur knot exterior X
2 Volume = 3.66386237670888
3 Isometric to census manifold: [m137(0,0)]
4 Cusp shape = [3.250000000000000 + 2.250000000000000*I]
5
6 Is X achiral? False
7 Order of isometry group = 2
8 Isometry group = Z/2
9 Self-isometries are: [0 -> 0
10 [1 0]
11 [0 1]
12 Extends to link, 0 -> 0
13 [-1 0]
14 [ 0 -1]
15 Extends to link]
16
17 Isometries X -> m137 are: [0 -> 0
18 [-1 -3]
19 [ 0 -1]
20 Extends to link, 0 -> 0

```

```

21 [1 3]
22 [0 1]
23 Extends to link]

```

Consequently:

- X is hyperbolic with $\text{vol}(X) \approx 3.66386$
- X is isometric to the 1-cusped census manifold $m137$
- The cusp shape of X is $\tau_X \approx \frac{13}{4} + \frac{9}{4}i$
- X admits no orientation reversing self-isometry
- The isometry group of X has order 2

SnapPy indicates the action of an isometry on cusps using 2×2 integer matrices. Each domain and target cusp torus must be equipped with an ordered homology basis $[\mu_D, \lambda_D]$ and $[\mu_T, \lambda_T]$ respectively. The matrix $A = \begin{bmatrix} a & c \\ b & d \end{bmatrix}$ implies that

$$x\mu_D + y\lambda_D \mapsto (ax + cy)\mu_T + (bx + dy)\lambda_T$$

In other words, the first column of A is the image of μ_D and the second column is the image of λ_D . If a manifold is defined in SnapPy as a link complement, then its cusps are equipped with standard topological meridian–longitude bases. This applies to X and Y , so ∂X is given the basis $[\mu_K, \lambda_K]$ and ∂Y is given the basis $[\mu_J, \lambda_J]$. Otherwise, SnapPy has its own heuristics for choosing bases of cusp tori. This applies to $m137$ and $t12072$.

From these conventions and the above computation we conclude:

- Every self-isometry of X acts trivially on slopes in ∂X
- Every isometry $X \rightarrow m137$ acts on slopes by $(p, q) \mapsto (p + 3q, q)$
- (Basis change) $X(k, 1)$ is homeomorphic to $m137(k + 3, 1)$ for each $k \in \mathbb{Z}$

We compute basic information about Y .

```

1 import snappy
2 # Jester knot exterior Y
3 Y = snappy.Manifold('DT: [(10,-20,-16,24,-14,22,18,-8,-4),(6,12,-2)], '
4     '[1,1,0,1,0,0,1,1,1,0,1,0]')
5 Y.dehn_fill((0,1),1)
6 Y = Y.filled_triangulation()
7 Yid = Y.identify()
8 IsomY = Y.symmetry_group()
9 print("Jester knot exterior Y")
10 print("Volume =", Y.volume())
11 print("Isometric to census manifold:", Yid)
12 print("Cusp shape =", Y.cusp_info('shape'))
13 print("")
14 print("Is Y achiral?", IsomY.is_amphicheiral())
15 print("Order of isometry group =", IsomY.order())
16 print("Isometry group =", IsomY)
17 print("Self-isometries are:", IsomY.isometries(), "\n")

```

```

18 print("Isometries Y -> t12072 are:", Y.is_isometric_to(snappy.Manifold('t12072'),
19       return_isometries=True))

```

```

1  Jester knot exterior Y
2  Volume = 7.32772475341775
3  Isometric to census manifold: [t12072(0,0)]
4  Cusp shape = [6.500000000000000 + 4.500000000000001*I]
5
6  Is Y achiral? False
7  Order of isometry group = 4
8  Isometry group = Z/2 + Z/2
9  Self-isometries are: [0 -> 0
10 [1 0]
11 [0 1]
12 Extends to link, 0 -> 0
13 [1 0]
14 [0 1]
15 Extends to link, 0 -> 0
16 [-1 0]
17 [ 0 -1]
18 Extends to link, 0 -> 0
19 [-1 0]
20 [ 0 -1]
21 Extends to link]
22
23 Isometries Y -> t12072 are: [0 -> 0
24 [1 6]
25 [0 1]
26 Extends to link, 0 -> 0
27 [-1 -6]
28 [ 0 -1]
29 Extends to link, 0 -> 0
30 [1 6]
31 [0 1]
32 Extends to link, 0 -> 0
33 [-1 -6]
34 [ 0 -1]
35 Extends to link]

```

Consequently:

- Y is hyperbolic with $\text{vol}(Y) \approx 7.32772$
- Y is isometric to the 1-cusped census manifold $t12072$
- The cusp shape of Y is $\tau_Y = \frac{13}{2} + \frac{9}{2}i$

- Y admits no orientation reversing self-isometry
- The isometry group of Y has order 4
- Every self-isometry of Y acts trivially on slopes in ∂Y
- Every isometry $Y \rightarrow t12072$ acts on slopes by $(p, q) \mapsto (p + 6q, q)$
- (Basis change) $Y(k, 1)$ is homeomorphic to $t12072(k + 6, 1)$ for each $k \in \mathbb{Z}$

Dunfield [Dun20] determined the exceptional slopes for SnapPy's census manifolds. Exceptional slopes are precisely those yielding nonhyperbolic fillings. For $m137$, the exceptional slopes are

- $(1, 0) \leftrightarrow 1/0 = \infty$ giving $S^1 \times S^2$ upon filling
- $(s, 1)$ where $s \in \{-2, -1, 0, 1\}$

By the basis change, the exceptional slopes of X are

- $(1, 0) \leftrightarrow 1/0 = \infty$ giving $S^1 \times S^2$ upon filling
- $(k, 1)$ where $k \in \{-5, -4, -3, -2\}$

So, the integral exceptional slopes of X are exactly $(k, 1)$ where $k \in \{-5, -4, -3, -2\}$.

For $t12072$, there is a unique exceptional slope

- $(1, 0) \leftrightarrow 1/0 = \infty$ giving $S^1 \times S^2$ upon filling

By the basis change, the exceptional slope of Y is

- $(1, 0) \leftrightarrow 1/0 = \infty$ giving $S^1 \times S^2$ upon filling

So, Y has no integral exceptional slope.

3. EXCEPTIONAL FILLINGS OF THE MAZUR KNOT EXTERIOR

We distinguish the four exceptional fillings $X(k, 1)$, where $k \in \{-5, -4, -3, -2\}$, pairwise up to homeomorphism. To do so, we count equivalence classes of epimorphisms from the fundamental groups $\pi_1(X(k, 1))$ to alternating groups A_n . The GAP command `GQuotients( $G, H$ )` returns a transversal for the equivalence relation on epimorphisms $G \twoheadrightarrow H$ defined by $f_1 \sim f_2$ if and only if there exists an automorphism ϕ of H such that $\phi \circ f_1 = f_2$.

```

1 import snappy
2 from sage.all import gap
3 # Distinguish exceptional fillings of X
4 X = snappy.Manifold('DT: [(14,12,-16,-10,18,-6),(-4,2,8)], [1,1,0,0,0,1,1,0,1]')
5 X.dehn_fill((0,1), 1)
6 X = X.filled_triangulation()
7 for n in range(5, 8):
8     H = gap(f"AlternatingGroup({n})")
9     print(f"Epimorphisms pi_1(X(k,1)) ->> A{n}")
10    for k in range(-5, -1):
11        CX = X.copy()
12        CX.dehn_fill((k,1), 0)
13        G = gap(CX.fundamental_group())
14        Q = gap.GQuotients(G, H)
15        print(f" X({k},1): {len(Q)}")

```

```

1 Epimorphisms pi_1(X(k,1)) ->> A5
2   X(-5,1): 1
3   X(-4,1): 1
4   X(-3,1): 0
5   X(-2,1): 0
6 Epimorphisms pi_1(X(k,1)) ->> A6
7   X(-5,1): 0
8   X(-4,1): 2
9   X(-3,1): 0
10  X(-2,1): 0
11 Epimorphisms pi_1(X(k,1)) ->> A7
12  X(-5,1): 0
13  X(-4,1): 5
14  X(-3,1): 2
15  X(-2,1): 0

```

Table 1 collects the results of this computation. Consequently, the fundamental groups $\pi_1(X(k,1))$ are pairwise nonisomorphic for $k \in \{-5, -4, -3, -2\}$. Therefore, the filled manifolds $X(k,1)$ are pairwise nonhomeomorphic for $k \in \{-5, -4, -3, -2\}$.

	A_5	A_6	A_7
$X(-5,1)$	1	0	0
$X(-4,1)$	1	2	5
$X(-3,1)$	0	0	2
$X(-2,1)$	0	0	0

TABLE 1. Numbers of classes of epimorphisms $\pi_1(X(k,1)) \twoheadrightarrow A_n$.

4. SYSTOLIC GEODESICS

We find all systolic geodesics in the filled manifolds $X(k,1)$ and $Y(k,1)$ for $k \in \mathbb{Z}$. We use a quantitative version of Thurston's hyperbolic Dehn surgery theorem for sufficiently large $|k|$. We cover the remaining finite list of values of k by direct computation.

The quantitative version of Thurston's hyperbolic Dehn surgery theorem that we use requires both the cusp shape and the systole of X and of Y . We computed the cusp shape above. We compute the systole using *m137* and *t12072* since SnapPy seems to perform better with difficult computations using census manifolds. As X and *m137* are isometric, their systoles are equal, and similarly for Y and *t12072*.

```

1 import snappy
2 # Systole of m137 and t12072
3 for name in ["m137", "t12072"]:
4     M = snappy.Manifold(name)
5     g = M.length_spectrum_alt(count=1, verified=True, bits_prec=100)[0]

```

```
6     print(f"sys({name}) =", g.length.real())
```

```
1     sys(m137) = 0.5097911466811016354624?
```

```
2     sys(t12072) = 1.01958229336220327?
```

Consequently, the verified systoles are

$$\text{sys}(X) = \text{sys}(m137) = 0.50979\dots$$

$$\text{sys}(Y) = \text{sys}(t12072) = 1.01958\dots$$

A trailing ? in SnapPy's output indicates that the final digit is uncertain while the preceding digits are verified correct. That yields a rigorous small interval containing the actual value.

Since both systoles exceed 0.145, the hypotheses of Futer–Purcell–Schleimer [FPS25, Thm. 3.19] and [FPS25, Thm. 3.23] reduce to requiring only that the normalized slope length be at least 9.97. Using the cusp shapes

$$\tau_X = \frac{13}{4} + \frac{9}{4}i \quad \text{and} \quad \tau_Y = \frac{13}{2} + \frac{9}{2}i$$

one computes that the normalized length of the filling slope $(k, 1)$ is at least 9.97 whenever $|k| \geq 19$ for X and $|k| \geq 28$ for Y . Therefore, by [FPS25, Thm. 3.23], the core of the filling is the unique systolic geodesic in $X(k, 1)$ for $|k| \geq 19$ and in $Y(k, 1)$ for $|k| \geq 28$.

Now, we check whether the core of the filling is the unique systolic geodesic for the remaining finite list of manifolds $X(k, 1)$ with $|k| \leq 18$. We perform the check using $m137$ and translate the results to X using the basis change. We must check the list $m137(s, 1)$ for $-15 \leq s \leq 21$. We compute the larger list $|s| \leq 25$ for overlap.

For $m137$, the ordinary routine `length_spectrum` occasionally succeeds when the verified routine `length_spectrum_alt` cannot certify the hyperbolic structure. Accordingly, we first attempt `length_spectrum` and fall back to `length_spectrum_alt`. For $t12072$, the verified routine succeeds throughout, so only `length_spectrum_alt` is needed.

```
1     import snappy
2     # Compare the filling core with the systole and test uniqueness for m137(s,1)
3     print(f"{'s':^3} {'core':^12} {'systole':^12}"
4           f"{'core=systole?':^15} {'unique?':^8}")
5     print("-" * 56)
6     for s in range(-25, 26):
7         if s in {-2, -1, 0, 1}:
8             print(f"{s:3d} {'----':>12} {'----':>12} {'nonhyperbolic':>15}")
9             continue
10        M = snappy.Manifold(f"m137({s},1)")
11        core = M.cusp_info(0).core_length.real()
12        try:
13            LS = M.length_spectrum(1.5, include_words=True)
14            systole = LS[0].length.real()
15            unique = abs(LS[0].length.real() - LS[1].length.real()) > 1e-10
16        except RuntimeError:
```



```

17     LS = M.length_spectrum_alt(count=4, verified=True, bits_prec=200)
18     systole = LS[0].length.real().center()
19     ordinary = [g for g in LS if g.core_curve is None]
20     unique = ordinary[0].length.real().upper() < \
21         ordinary[1].length.real().lower()
22     core_is_systole = abs(core - systole) < 1e-10
23     print(f"{s:3d} {core:12.8f} {systole:12.8f}"
24           f"{str(core_is_systole):^15} {str(unique):^8}")

```

	s	core	systole	core=systole?	unique?
2	-----				
3	-25	0.02286701	0.02286701	True	True
4	-24	0.02481388	0.02481388	True	True
5	-23	0.02701946	0.02701946	True	True
6	-22	0.02953154	0.02953154	True	True
7	-21	0.03240940	0.03240940	True	True
8	-20	0.03572732	0.03572732	True	True
9	-19	0.03957934	0.03957934	True	True
10	-18	0.04408588	0.04408588	True	True
11	-17	0.04940311	0.04940311	True	True
12	-16	0.05573632	0.05573632	True	True
13	-15	0.06335964	0.06335964	True	True
14	-14	0.07264532	0.07264532	True	True
15	-13	0.08410857	0.08410857	True	True
16	-12	0.09847779	0.09847779	True	True
17	-11	0.11680747	0.11680747	True	True
18	-10	0.14066482	0.14066482	True	True
19	-9	0.17244646	0.17244646	True	True
20	-8	0.21592633	0.21592633	True	True
21	-7	0.27720467	0.27720467	True	True
22	-6	0.36629268	0.36629268	True	True
23	-5	0.49966077	0.49966077	True	True
24	-4	0.70647343	0.59995705	False	True
25	-3	1.06960599	0.49916351	False	True
26	-2	---	---	nonhyperbolic	
27	-1	---	---	nonhyperbolic	
28	0	---	---	nonhyperbolic	
29	1	---	---	nonhyperbolic	
30	2	1.43906664	0.36613070	False	True
31	3	0.85704045	0.56397216	False	True
32	4	0.59110354	0.59110354	True	True
33	5	0.42605455	0.42605455	True	True
34	6	0.31743363	0.31743363	True	True

35	7	0.24383067	0.24383067	True	True
36	8	0.19241222	0.19241222	True	True
37	9	0.15537015	0.15537015	True	True
38	10	0.12791980	0.12791980	True	True
39	11	0.10706453	0.10706453	True	True
40	12	0.09087339	0.09087339	True	True
41	13	0.07806528	0.07806528	True	True
42	14	0.06776650	0.06776650	True	True
43	15	0.05936618	0.05936618	True	True
44	16	0.05242748	0.05242748	True	True
45	17	0.04663167	0.04663167	True	True
46	18	0.04174201	0.04174201	True	True
47	19	0.03757975	0.03757975	True	True
48	20	0.03400801	0.03400801	True	True
49	21	0.03092051	0.03092051	True	True
50	22	0.02823379	0.02823379	True	True
51	23	0.02588158	0.02588158	True	True
52	24	0.02381071	0.02381071	True	True
53	25	0.02197816	0.02197816	True	True

Consequently, the following hold for $m137(s, 1)$:

- For $s \notin \{-4, -3, \dots, 3\}$, the core of the filling is the unique systolic geodesic
- For $s \in \{-4, -3, 2, 3\}$, the systolic geodesic is not the core of the filling but is unique
- $s \in \{-2, -1, 0, 1\}$ are the exceptional slopes

Therefore, the following hold for $X(k, 1)$:

- For $k \notin \{-7, -6, \dots, 0\}$, the core of the filling is the unique systolic geodesic
- For $k \in \{-7, -6, -1, 0\}$, the systolic geodesic is not the core of the filling but is unique
- $k \in \{-5, -4, -3, -2\}$ are the exceptional slopes

Now, we check whether the core of the filling is the unique systolic geodesic for the remaining finite list of manifolds $Y(k, 1)$ with $|k| \leq 27$. We perform the check using $t12072$ and translate the results to Y using the basis change. We must check the list $t12072(s, 1)$ for $-21 \leq s \leq 33$. We compute the larger list $|s| \leq 35$ for overlap.

```

1 import snappy
2 # Compare the filling core with the systole and test uniqueness for t12072(s,1)
3 print(f"{'s':^3} {'core':^12} {'systole':^12}"
4       f"{'core=systole?':^15} {'unique?':^8}")
5 print("-" * 56)
6 for s in range(-35, 36):
7     M = snappy.Manifold(f"t12072({s},1)")
8     core = M.cusp_info(0).core_length.real()
9     LS = M.length_spectrum_alt(count=4, verified=True, bits_prec=200)
10    systole = LS[0].length.real()

```

```

11     core_is_systole = abs(core - systole.center()) < 1e-10
12     unique = systole.upper() < LS[1].length.real().lower()
13     print(f"{s:3d} {core:12.8f} {systole.center():12.8f}"
14           f"{str(core_is_systole):^15} {str(unique):^8}")

```

	s	core	systole	core=systole?	unique?
2	-----				
3	-35	0.02334634	0.02334634	True	True
4	-34	0.02473530	0.02473530	True	True
5	-33	0.02625116	0.02625116	True	True
6	-32	0.02790974	0.02790974	True	True
7	-31	0.02972941	0.02972941	True	True
8	-30	0.03173155	0.03173155	True	True
9	-29	0.03394118	0.03394118	True	True
10	-28	0.03638771	0.03638771	True	True
11	-27	0.03910591	0.03910591	True	True
12	-26	0.04213709	0.04213709	True	True
13	-25	0.04553065	0.04553065	True	True
14	-24	0.04934594	0.04934594	True	True
15	-23	0.05365487	0.05365487	True	True
16	-22	0.05854504	0.05854504	True	True
17	-21	0.06412408	0.06412408	True	True
18	-20	0.07052518	0.07052518	True	True
19	-19	0.07791465	0.07791465	True	True
20	-18	0.08650197	0.08650197	True	True
21	-17	0.09655357	0.09655357	True	True
22	-16	0.10841174	0.10841174	True	True
23	-15	0.12252094	0.12252094	True	True
24	-14	0.13946474	0.13946474	True	True
25	-13	0.16001802	0.16001802	True	True
26	-12	0.18522130	0.18522130	True	True
27	-11	0.21648661	0.21648661	True	True
28	-10	0.25574812	0.25574812	True	True
29	-9	0.30567411	0.30567411	True	True
30	-8	0.36996287	0.36996287	True	True
31	-7	0.45375527	0.45375527	True	True
32	-6	0.56420104	0.56420104	True	True
33	-5	0.71087754	0.71087754	True	True
34	-4	0.90188600	0.90188600	True	True
35	-3	1.11171893	0.77858590	False	True
36	-2	1.24807978	0.67096709	False	True
37	-1	1.29436872	0.61977976	False	True
38	0	1.29436872	0.61870851	False	True

39	1	1.24807978	0.66785203	False	True
40	2	1.11171893	0.77442060	False	True
41	3	0.90188600	0.90188600	True	False
42	4	0.71087754	0.71087754	True	True
43	5	0.56420104	0.56420104	True	True
44	6	0.45375527	0.45375527	True	True
45	7	0.36996287	0.36996287	True	True
46	8	0.30567411	0.30567411	True	True
47	9	0.25574812	0.25574812	True	True
48	10	0.21648661	0.21648661	True	True
49	11	0.18522130	0.18522130	True	True
50	12	0.16001802	0.16001802	True	True
51	13	0.13946474	0.13946474	True	True
52	14	0.12252094	0.12252094	True	True
53	15	0.10841174	0.10841174	True	True
54	16	0.09655357	0.09655357	True	True
55	17	0.08650197	0.08650197	True	True
56	18	0.07791465	0.07791465	True	True
57	19	0.07052518	0.07052518	True	True
58	20	0.06412408	0.06412408	True	True
59	21	0.05854504	0.05854504	True	True
60	22	0.05365487	0.05365487	True	True
61	23	0.04934594	0.04934594	True	True
62	24	0.04553065	0.04553065	True	True
63	25	0.04213709	0.04213709	True	True
64	26	0.03910591	0.03910591	True	True
65	27	0.03638771	0.03638771	True	True
66	28	0.03394118	0.03394118	True	True
67	29	0.03173155	0.03173155	True	True
68	30	0.02972941	0.02972941	True	True
69	31	0.02790974	0.02790974	True	True
70	32	0.02625116	0.02625116	True	True
71	33	0.02473530	0.02473530	True	True
72	34	0.02334634	0.02334634	True	True
73	35	0.02207058	0.02207058	True	True

Consequently, the following hold for $t12072(s, 1)$:

- For $s \notin \{-3, -2, \dots, 3\}$, the core of the filling is the unique systolic geodesic
- For $s \in \{-3, -2, \dots, 2\}$, the systolic geodesic is not the core of the filling but is unique
- For $s = 3$, the core of the filling is a systolic geodesic but it is not unique

Therefore, the following hold for $Y(k, 1)$:

- For $k \notin \{-9, -8, \dots, -3\}$, the core of the filling is the unique systolic geodesic

- For $k \in \{-9, -8, \dots, -4\}$, the systolic geodesic is not the core of the filling but is unique
- For $k = -3$, the core of the filling is a systolic geodesic but it is not unique

5. DRILLING SYSTOLIC GEODESICS

Drilling unique systolic geodesics is a powerful method for producing topological invariants. We drill the unique noncore systolic geodesics for the hyperbolic filled manifolds $X(k, 1)$ and $Y(k, 1)$. For $Y(-3, 1)$, we show there are exactly two systolic geodesics and drill each of them. We compute with $m137(s, 1)$ and $t12072(s, 1)$ and translate the results to $X(k, 1)$ and $Y(k, 1)$ using the basis changes.

```

1 import snappy
2 # Drill the unique systolic geodesic for the remaining
3 # hyperbolic fillings of m137(s,1)
4 print(f"{'s':^3} {'systole':^12} {'systole drilled manifold'}")
5 print("-" * 82)
6 for s in [-4, -3, 2, 3]:
7     M = snappy.Manifold(f"m137({s},1)")
8     D = M.drill(0)
9     D.dehn_fill((1,0),1)
10    F = D.filled_triangulation([0])
11    g = F.length_spectrum_alt(count=1, verified=True, bits_prec=200)[0]
12    H = F.drill_word(g.word, verified=True, bits_prec=200)
13    print(f"{s:3d} {g.length.real():12.8f} {H.identify()}")

```

s	systole	systole drilled manifold
-4	0.59995705	[m081(0,0)]
-3	0.49916351	[m011(0,0)]
2	0.36613070	[m004(0,0), 4_1(0,0), K2_1(0,0), K4a1(0,0), otet02_00001(0,0)]
3	0.56397216	[m038(0,0)]

```

1 import snappy
2 # Drill the unique systolic geodesic for the remaining
3 # hyperbolic fillings of t12072(s,1)
4 print(f"{'s':^3} {'systole':^12} {'systole drilled manifold'}")
5 print("-" * 67)
6 for s in [-3, -2, -1, 0, 1, 2]:
7     M = snappy.Manifold(f"t12072({s},1)")
8     g = M.length_spectrum(2.0, include_words=True)[0]
9     H = M.drill_word(g.word)
10    print(f"{s:3d} {g.length.real():12.8f} {H.identify()}")

```

s	systole	systole drilled manifold
-3	0.77858590	[v3519(0,0)]
-2	0.67096709	[v3200(0,0)]
-1	0.61977976	[v2919(0,0)]
0	0.61870851	[v2911(0,0)]
1	0.66785203	[v3184(0,0)]
2	0.77442060	[v3505(0,0), 8_21(0,0), K7_127(0,0), K8n2(0,0)]

Consequently, we obtain the drilled manifolds in Table 2.

$X(k, 1)$		$Y(k, 1)$	
k	Systole-drilled manifold	k	Systole-drilled manifold
-7	$m081$	-9	$v3519$
-6	$m011$	-8	$v3200$
-5, ..., -2	exceptional slopes	-7	$v2919$
-1	$m004$	-6	$v2911$
0	$m038$	-5	$v3184$
otherwise	$m137$	-4	$v3505$
		-3	not unique
		otherwise	$t12072$

TABLE 2. Manifolds obtained by drilling the unique systolic geodesic in $X(k, 1)$ and $Y(k, 1)$.

Now, we show that $Y(-3, 1)$ contains exactly two systolic geodesics and we drill each of them. We thank Nathan Dunfield for this code.

```

1 import snappy
2 # Show that t12072(3,1) has exactly two systolic geodesics
3 # and that a self-isometry interchanges them
4 M = snappy.Manifold("t12072(3,1)")
5 S = M.symmetric_triangulation()
6 print("Verified beginning of the length spectrum of t12072(3,1):")
7 LS = S.length_spectrum_alt(count=4, verified=True, bits_prec=1000)
8 for g in LS:
9     print(g)
10 print()
11 print("Drilling each systolic geodesic:")
12 for g in LS[:2]:
13     T = S.copy()
14     U = T.drill_word(g.word, verified=True, bits_prec=1000)
15     print(f"{g.word:<20} {U.identify()}")

```

```

16 # Drill both systolic geodesics, yielding a 2-cusped hyperbolic 3-manifold.
17 S.dehn_fill([(0,0),(0,0)])
18 print()
19 print("Self-isometries of the drilled manifold:")
20 for iso in S.is_isometric_to(S, return_isometries=True):
21     print(iso)

```

```

1 Verified beginning of the length spectrum of t12072(3,1):
2 Length                                     Core curve Word
3 0.90188599874433... - 0.68042658939886...*I Cusp 1      fGGGGGG
4 0.90188599874433... - 0.68042658939886...*I Cusp 0      bcgdbCF
5 1.43277618896090... - 2.84374651337138...*I -            dE
6 1.61363312906004... - 2.06618680610597...*I -            aBEGCgcADGCgCgFbe
7 1.61363312906004... - 2.06618680610597...*I -            aBFggfcADe
8
9 Drilling each systolic geodesic:
10 fGGGGGG [t12072(0,0)]
11 bcgdbCF [t12072(0,0)]
12
13 Self-isometries of the drilled manifold:
14 0 -> 1   1 -> 0
15 [-1 -1] [-1  1]
16 [ 0 -1] [ 0 -1]
17 Extends to link
18 0 -> 0   1 -> 1
19 [1  0]  [1  0]
20 [0  1]  [0  1]
21 Extends to link
22 0 -> 0   1 -> 1
23 [-1  0] [-1  0]
24 [ 0 -1] [ 0 -1]
25 Extends to link
26 0 -> 1   1 -> 0
27 [1  1]  [1 -1]
28 [0  1]  [0  1]
29 Extends to link
30 0 -> 0   1 -> 1
31 [-1  0] [-1  0]
32 [ 0 -1] [ 0 -1]
33 Extends to link
34 0 -> 1   1 -> 0
35 [1  1]  [1 -1]
36 [0  1]  [0  1]

```

```

37 Extends to link
38 0 -> 1    1 -> 0
39 [-1 -1]   [-1  1]
40 [ 0 -1]   [ 0 -1]
41 Extends to link
42 0 -> 0    1 -> 1
43 [1  0]    [1  0]
44 [0  1]    [0  1]
45 Extends to link

```

The verified length spectrum of $t12072(3,1)$ begins with two geodesics, Cusp 1 and Cusp 0, with apparently equal lengths. Drilling each of them yields $t12072$. Drilling both of them yields a two-cusped manifold with a self-isometry—the first isometry listed—interchanging the cusps and preserving the meridians up to orientation. Consequently, $t12072(3,1)$ has a self-isometry interchanging the geodesics Cusp 1 and Cusp 0. Therefore, Cusp 1 and Cusp 0 are the two systolic geodesics of $t12072(3,1)$. It follows that $Y(-3,1)$ contains exactly two systolic geodesics and drilling each of them yields $t12072$.

We verify that the 12 drilled manifolds are pairwise nonhomeomorphic. By Mostow–Prasad rigidity, it suffices to show they are pairwise nonisometric.

```

1 import snappy
2 # Verify the 12 systole drilled manifolds are distinct
3 names = ["m137", "t12072", "m081", "m011", "m004", "m038",
4          "v3519", "v3200", "v2919", "v2911", "v3184", "v3505"]
5 manifolds = [snappy.Manifold(name) for name in names]
6 print("Checking pairwise isometry...")
7 duplicate = False
8 for i in range(len(manifolds)):
9     for j in range(i + 1, len(manifolds)):
10         if manifolds[i].is_isometric_to(manifolds[j]):
11             duplicate = True
12             print(f"{names[i]} is isometric to {names[j]}")
13 if not duplicate:
14     print("All 12 manifolds are pairwise nonisometric.")

```

```

1 Checking pairwise isometry...
2 All 12 manifolds are pairwise nonisometric.

```

6. VOLUMES

We perform volume computations on the hyperbolic filled manifolds $X(k,1)$ and $Y(k,1)$ which suggest our conjectures on volume monotonicity and symmetry.

```

1 import snappy

```



```

2  # Volumes of Dehn fillings  $X(k,1)$  of  $X$ 
3  X = snappy.Manifold('DT: [(14,12,-16,-10,18,-6),(-4,2,8)], [1,1,0,0,0,1,1,0,1]')
4  X.dehn_fill((0,1),1)
5  X = X.filled_triangulation()
6  Xid = X.identify()
7  print("Mazur knot exterior X")
8  print("Volume =", X.volume())
9  print("Isometric to census manifold:", Xid, "\n")
10 M = snappy.Manifold("m137")
11 print("Volumes of some integral Dehn fillings:\n")
12 print(f'{"k":^5} {"vol(X(k,1r))":>15} {"vol(m137(k+3r,1r))":>15} {"equal?":>7}')
13 print("-" * 50)
14 k_values = list(range(-1000, -994)) + list(range(-20, 21)) + list(range(994, 1001))
15 for k in k_values:
16     if k in {-5, -4, -3, -2}:
17         print(f'{"k:5d} {"nonhyperbolic":>15}')
18         continue
19     DFX = X.copy()
20     DFX.dehn_fill((k,1),0)
21     vDFX = DFX.volume()
22     DFM = M.copy()
23     DFM.dehn_fill((k+3,1),0)
24     vDFM = DFM.volume()
25     equal = abs(vDFX - vDFM) < 1e-10
26     print(f'{"k:5d} {"vDFX:15.10f} {"vDFM:15.10f} {"str(equal):>7}')
27 print()
28 print("Verify vol(X(k,1)) strictly increases as k increases from -1 to 1000:")
29 flag = True
30 X1 = X.copy()
31 X1.dehn_fill((-1,1),0)
32 v1 = X1.volume()
33 for k in range(0,1001):
34     X2 = X.copy()
35     X2.dehn_fill((k,1),0)
36     v2 = X2.volume()
37     #print(f'{"k} {"v1} {"v2}')
38     if v1 < v2:
39         v1 = v2
40         continue
41     else:
42         flag = False
43         print(f"Volume stopped increasing at k = {k}")
44         break

```

```

45 print(f"{flag}")
46 print()
47 print("Verify vol(X(k,1)) strictly increases as k decreases from -6 to -1000:")
48 flag = True
49 X1 = X.copy()
50 X1.dehn_fill((-6,1),0)
51 v1 = X1.volume()
52 for k in range(-7,-1001,-1):
53     X2 = X.copy()
54     X2.dehn_fill((k,1),0)
55     v2 = X2.volume()
56     #print(f"{k} {v1} {v2}")
57     if v1 < v2:
58         v1 = v2
59         continue
60     else:
61         flag = False
62         print(f"Volume stopped increasing at k = {k}")
63         break
64 print(f"{flag}")

```

```

1 Mazur knot exterior X
2 Volume = 3.66386237670888
3 Isometric to census manifold: [m137(0,0)]
4
5 Volumes of some integral Dehn fillings:
6
7   k          vol(X(k,1))  vol(m137(k+3,1))  equal?
8   -----
9  -1000      3.6638400252    3.6638400252    True
10  -999       3.6638399803    3.6638399803    True
11  -998       3.6638399352    3.6638399352    True
12  -997       3.6638398900    3.6638398900    True
13  -996       3.6638398447    3.6638398447    True
14  -995       3.6638397992    3.6638397992    True
15  -20        3.5861901213    3.5861901213    True
16  -19        3.5762257574    3.5762257574    True
17  -18        3.5642301473    3.5642301473    True
18  -17        3.5496168070    3.5496168070    True
19  -16        3.5315737988    3.5315737988    True
20  -15        3.5089527264    3.5089527264    True
21  -14        3.4800896428    3.4800896428    True
22  -13        3.4425070875    3.4425070875    True

```

23	-12	3.3924017384	3.3924017384	True
24	-11	3.3237331543	3.3237331543	True
25	-10	3.2265517707	3.2265517707	True
26	-9	3.0838610497	3.0838610497	True
27	-8	2.8656302333	2.8656302333	True
28	-7	2.5162213804	2.5162213804	True
29	-6	1.9122102501	1.9122102501	True
30	-5	nonhyperbolic		
31	-4	nonhyperbolic		
32	-3	nonhyperbolic		
33	-2	nonhyperbolic		
34	-1	1.3985088842	1.3985088842	True
35	0	2.2597671326	2.2597671326	True
36	1	2.7124588084	2.7124588084	True
37	2	2.9868370451	2.9868370451	True
38	3	3.1623672864	3.1623672864	True
39	4	3.2795522103	3.2795522103	True
40	5	3.3608908541	3.3608908541	True
41	6	3.4193302144	3.4193302144	True
42	7	3.4625868297	3.4625868297	True
43	8	3.4954325756	3.4954325756	True
44	9	3.5209246919	3.5209246919	True
45	10	3.5410862256	3.5410862256	True
46	11	3.5572951021	3.5572951021	True
47	12	3.5705142075	3.5705142075	True
48	13	3.5814318669	3.5814318669	True
49	14	3.5905501777	3.5905501777	True
50	15	3.5982420594	3.5982420594	True
51	16	3.6047890027	3.6047890027	True
52	17	3.6104066018	3.6104066018	True
53	18	3.6152621701	3.6152621701	True
54	19	3.6194871148	3.6194871148	True
55	20	3.6231857749	3.6231857749	True
56	994	3.6638400476	3.6638400476	True
57	995	3.6638400923	3.6638400923	True
58	996	3.6638401369	3.6638401369	True
59	997	3.6638401813	3.6638401813	True
60	998	3.6638402256	3.6638402256	True
61	999	3.6638402698	3.6638402698	True
62	1000	3.6638403139	3.6638403139	True

63

64 Verify $\text{vol}(X(k,1))$ strictly increases as k increases from -1 to 1000:

65 True

```

66
67 Verify vol( $X(k,1)$ ) strictly increases as  $k$  decreases from  $-6$  to  $-1000$ :
68 True

```

Consequently, we conjecture that:

- $\text{vol}(X(k,1))$ strictly increases as k increases beginning at -1
 - $\text{vol}(X(k,1))$ strictly increases as k decreases beginning at -6
 - $X(-1,1)$ has the minimum volume among all hyperbolic $X(k,1)$
-

```

1  import snappy
2  # Volumes of Dehn fillings  $Y(k,1)$  of  $Y$ 
3  Y = snappy.Manifold('DT: [(10,-20,-16,24,-14,22,18,-8,-4),(6,12,-2)], '
4      '[1,1,0,1,0,0,1,1,1,0,1,0]')
5  Y.dehn_fill((0,1),1)
6  Y = Y.filled_triangulation()
7  Yid = Y.identify()
8  print("Jester knot exterior Y")
9  print("Volume =", Y.volume())
10 print("Isometric to census manifold:", Yid, "\n")
11 T = snappy.Manifold("t12072")
12 print("Volumes of some integral Dehn fillings:\n")
13 print(f"{'k':^5} {'vol(Y(k,1r))':>15} {'vol(t12072(k+6r,1r))':>18} {'equal?':>7}")
14 print("-" * 53)
15 k_values = list(range(-1000, -994)) + list(range(-20, 21)) + list(range(994, 1001))
16 for k in k_values:
17     DFY = Y.copy()
18     DFY.dehn_fill((k,1),0)
19     vDFY = DFY.volume()
20     DFT = T.copy()
21     DFT.dehn_fill((k+6,1),0)
22     vDFT = DFT.volume()
23     equal = abs(vDFY - vDFT) < 1e-10
24     print(f"{'k':5d} {'vDFY':15.10f} {'vDFT':18.10f} {'str(equal):>7}")
25 print()
26 print("Verify vol( $Y(k,1)$ ) strictly increases as  $k$  increases from  $-6$  to  $1000$ :")
27 flag = True
28 Y1 = Y.copy()
29 Y1.dehn_fill((-6,1),0)
30 v1 = Y1.volume()
31 for k in range(-5,1001):
32     Y2 = Y.copy()
33     Y2.dehn_fill((k,1),0)
34     v2 = Y2.volume()
35     #print(f"{'k'} {'v1'} {'v2'}")

```

```

36     if v1 < v2:
37         v1 = v2
38         continue
39     else:
40         flag = False
41         print(f"Volume stopped increasing at k = {k}")
42         break
43 print(f"{flag}")
44 print()
45 print("Verify vol(Y(k,1)) strictly increases as k decreases from -7 to -1000:")
46 flag = True
47 Y1 = Y.copy()
48 Y1.dehn_fill((-7,1),0)
49 v1 = Y1.volume()
50 for k in range(-8,-1001,-1):
51     Y2 = Y.copy()
52     Y2.dehn_fill((k,1),0)
53     v2 = Y2.volume()
54     #print(f"{k} {v1} {v2}")
55     if v1 < v2:
56         v1 = v2
57         continue
58     else:
59         flag = False
60         print(f"Volume stopped increasing at k = {k}")
61         break
62 print(f"{flag}")

```

```

1 Jester knot exterior Y
2 Volume = 7.32772475341775
3 Isometric to census manifold: [t12072(0,0)]
4
5 Volumes of some integral Dehn fillings:
6
7   k          vol(Y(k,1))  vol(t12072(k+6,1))  equal?
8   -----
9  -1000      7.3276797581      7.3276797581      True
10  -999       7.3276796674      7.3276796674      True
11  -998       7.3276795764      7.3276795764      True
12  -997       7.3276794851      7.3276794851      True
13  -996       7.3276793936      7.3276793936      True
14  -995       7.3276793018      7.3276793018      True
15  -20        7.1085194314      7.1085194314      True

```

16	-19	7.0762146287	7.0762146287	True
17	-18	7.0366039004	7.0366039004	True
18	-17	6.9874700845	6.9874700845	True
19	-16	6.9257790105	6.9257790105	True
20	-15	6.8473580349	6.8473580349	True
21	-14	6.7464811852	6.7464811852	True
22	-13	6.6154126726	6.6154126726	True
23	-12	6.4441579271	6.4441579271	True
24	-11	6.2213963155	6.2213963155	True
25	-10	5.9406422204	5.9406422204	True
26	-9	5.6250329931	5.6250329931	True
27	-8	5.3589516115	5.3589516115	True
28	-7	5.2183624790	5.2183624790	True
29	-6	5.2183624790	5.2183624790	True
30	-5	5.3589516115	5.3589516115	True
31	-4	5.6250329931	5.6250329931	True
32	-3	5.9406422204	5.9406422204	True
33	-2	6.2213963155	6.2213963155	True
34	-1	6.4441579271	6.4441579271	True
35	0	6.6154126726	6.6154126726	True
36	1	6.7464811852	6.7464811852	True
37	2	6.8473580349	6.8473580349	True
38	3	6.9257790105	6.9257790105	True
39	4	6.9874700845	6.9874700845	True
40	5	7.0366039004	7.0366039004	True
41	6	7.0762146287	7.0762146287	True
42	7	7.1085194314	7.1085194314	True
43	8	7.1351524823	7.1351524823	True
44	9	7.1573309124	7.1573309124	True
45	10	7.1759714422	7.1759714422	True
46	11	7.1917723785	7.1917723785	True
47	12	7.2052715722	7.2052715722	True
48	13	7.2168877176	7.2168877176	True
49	14	7.2269500639	7.2269500639	True
50	15	7.2357200093	7.2357200093	True
51	16	7.2434069574	7.2434069574	True
52	17	7.2501800825	7.2501800825	True
53	18	7.2561771445	7.2561771445	True
54	19	7.2615111597	7.2615111597	True
55	20	7.2662754930	7.2662754930	True
56	994	7.3276803855	7.3276803855	True
57	995	7.3276804740	7.3276804740	True
58	996	7.3276805623	7.3276805623	True

```

59     997      7.3276806504      7.3276806504      True
60     998      7.3276807381      7.3276807381      True
61     999      7.3276808256      7.3276808256      True
62    1000      7.3276809129      7.3276809129      True
63
64 Verify vol(Y(k,1)) strictly increases as k increases from -6 to 1000:
65 True
66
67 Verify vol(Y(k,1)) strictly increases as k decreases from -7 to -1000:
68 True

```

Consequently, we conjecture that:

- $\text{vol}(Y(k,1))$ strictly increases as k increases beginning at -6
 - $\text{vol}(Y(k,1))$ strictly increases as k decreases beginning at -7
 - $Y(-6,1)$ and $Y(-7,1)$ have the minimum volume among all $Y(k,1)$
-

```

1  import snappy
2  # Jester volume symmetry vol(Y(k,1))=vol(Y(-k-13,1))
3  # vol(t12072(s,1))=vol(t12072(-s-1,1))
4  Y = snappy.Manifold('DT: [(10,-20,-16,24,-14,22,18,-8,-4),(6,12,-2)], '
5      '[1,1,0,1,0,0,1,1,1,0,1,0]')
6  Y.dehn_fill((0,1),1)
7  Y = Y.filled_triangulation()
8  Yid = Y.identify()
9  print("Jester knot exterior Y")
10 print("Volume =", Y.volume())
11 print("Isometric to census manifold:", Yid, "\n")
12 #T = snappy.Manifold("t12072")
13 print("Volumes of some integral Dehn fillings:\n")
14 print(f"{'k':^5} {'vol(Y(k,1r))':>15} {'vol(Y(-k-13r,1r))':>18} {'equal?':>7}")
15 print("-" * 52)
16 k_values = list(range(-1000, -994)) + list(range(-20, 21)) + list(range(994, 1001))
17 for k in k_values:
18     Y1 = Y.copy()
19     Y1.dehn_fill((k,1),0)
20     v1 = Y1.volume()
21     Y2 = Y.copy()
22     Y2.dehn_fill((-k-13,1),0)
23     v2 = Y2.volume()
24     equal = abs(v1 - v2) < 1e-10
25     print(f"{'k':5d} {'v1:15.10f} {'v2:18.10f} {'str(equal):>7}")
26 print()
27 print("Verify vol(Y(k,1))=vol(Y(-k-13,1)) for -1000<=k<=1000:")
28 flag = True

```

```

29 for k in range(-1000,1001):
30     Y1 = Y.copy()
31     Y1.dehn_fill((k,1),0)
32     v1 = Y1.volume()
33     Y2 = Y.copy()
34     Y2.dehn_fill((-k-13,1),0)
35     v2 = Y2.volume()
36     equal = abs(v1 - v2) < 1e-10
37     #print(f"{k} {v1} {v2}")
38     if equal:
39         continue
40     else:
41         flag = False
42         print(f"Volumes unequal at k = {k}")
43         break
44 print(f"{flag}")

```

```

1 Jester knot exterior Y
2 Volume = 7.32772475341775
3 Isometric to census manifold: [t12072(0,0)]
4
5 Volumes of some integral Dehn fillings:
6
7   k          vol(Y(k,1))    vol(Y(-k-13,1))    equal?
8   -----
9  -1000      7.3276797581      7.3276797581      True
10  -999       7.3276796674      7.3276796674      True
11  -998       7.3276795764      7.3276795764      True
12  -997       7.3276794851      7.3276794851      True
13  -996       7.3276793936      7.3276793936      True
14  -995       7.3276793018      7.3276793018      True
15  -20        7.1085194314      7.1085194314      True
16  -19        7.0762146287      7.0762146287      True
17  -18        7.0366039004      7.0366039004      True
18  -17        6.9874700845      6.9874700845      True
19  -16        6.9257790105      6.9257790105      True
20  -15        6.8473580349      6.8473580349      True
21  -14        6.7464811852      6.7464811852      True
22  -13        6.6154126726      6.6154126726      True
23  -12        6.4441579271      6.4441579271      True
24  -11        6.2213963155      6.2213963155      True
25  -10        5.9406422204      5.9406422204      True
26  -9         5.6250329931      5.6250329931      True

```


27	-8	5.3589516115	5.3589516115	True
28	-7	5.2183624790	5.2183624790	True
29	-6	5.2183624790	5.2183624790	True
30	-5	5.3589516115	5.3589516115	True
31	-4	5.6250329931	5.6250329931	True
32	-3	5.9406422204	5.9406422204	True
33	-2	6.2213963155	6.2213963155	True
34	-1	6.4441579271	6.4441579271	True
35	0	6.6154126726	6.6154126726	True
36	1	6.7464811852	6.7464811852	True
37	2	6.8473580349	6.8473580349	True
38	3	6.9257790105	6.9257790105	True
39	4	6.9874700845	6.9874700845	True
40	5	7.0366039004	7.0366039004	True
41	6	7.0762146287	7.0762146287	True
42	7	7.1085194314	7.1085194314	True
43	8	7.1351524823	7.1351524823	True
44	9	7.1573309124	7.1573309124	True
45	10	7.1759714422	7.1759714422	True
46	11	7.1917723785	7.1917723785	True
47	12	7.2052715722	7.2052715722	True
48	13	7.2168877176	7.2168877176	True
49	14	7.2269500639	7.2269500639	True
50	15	7.2357200093	7.2357200093	True
51	16	7.2434069574	7.2434069574	True
52	17	7.2501800825	7.2501800825	True
53	18	7.2561771445	7.2561771445	True
54	19	7.2615111597	7.2615111597	True
55	20	7.2662754930	7.2662754930	True
56	994	7.3276803855	7.3276803855	True
57	995	7.3276804740	7.3276804740	True
58	996	7.3276805623	7.3276805623	True
59	997	7.3276806504	7.3276806504	True
60	998	7.3276807381	7.3276807381	True
61	999	7.3276808256	7.3276808256	True
62	1000	7.3276809129	7.3276809129	True
63				
64	Verify $\text{vol}(Y(k,1))=\text{vol}(Y(-k-13,1))$ for $-1000\leq k\leq 1000$:			
65	True			

Consequently, we conjecture that:

- $\text{vol}(Y(k,1)) = \text{vol}(Y(-k-13,1))$ for all $k \in \mathbb{Z}$

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DEPARTMENT OF MATHEMATICS, OBERLIN COLLEGE
Email address: `jcalcut@oberlin.edu`

DEPARTMENT OF MATHEMATICS, UNIVERSITY OF MICHIGAN
Email address: `yyadu@umich.edu`